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Effectiveness of Case-based Learning in Chinese Dental Education: A systematic review and meta-analysis

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Title: Effectiveness of Case-based Learning in Chinese Dental Education: A systematic review and meta-analysis

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Haitao Dong, Xinyuan Zhang and Xin Zhang carried out the entire procedure including the literature search, data extraction, performed the statistical analysis, drafted and revised the manuscript. **Jizhi Zhao and Xiaomin Wu** contributed to the revision of the manuscript. **Chunlan Guo and Lian Zhou** conceived of the study, coordinated and participated in the entire process of drafting and revising the manuscript.

Data Sharing Statement:

All data relevant to the study are included in the article or uploaded as online supplemental information. No unpublished data are available.

Ethical approval and informed consent

Since this is a systematic review, ethical approval and informed consent are not required.

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Abstract

Objective The application of case-based learning (CBL) pedagogy has been emerging as an improved and more practical learning method across the Chinese dental education system. This article provides a critical overview of CBL pedagogy and further assess the effectiveness of this teaching model in Chinese dental education.

Design A Systematic review and meta-analysis.

Methods Literatures published up to December 2019 were searched in the following electronic databases: China National Knowledge Infrastructure (CNKI), China Science Periodical Database (VIP), Chinese Biomedical documental database (CBM), Wanfang data, PubMed, EMBASE and Cochrane Central Register of Control Trials. All randomized controlled trials (RCTs) that compared the effectiveness of the CBL teaching model and the traditional lecture-based learning (LBL) model in all dental disciplines were included. The assessment of methodological quality was based on the guidelines described in the Cochrane Handbook for Systematic Reviews, and the meta-analysis was performed using the software RevMan v5.3

Results A total of 30 RCTs were finally included, with a total sample size of 2356 dental students. The CBL pedagogy significantly increased knowledges scores (SMD=1.53, 95% CI: 0.93-2.53, P<0.0001), skill scores (SMD =1.22, 95% CI: 0.34-2.11, P<0.0001), comprehensive ability scores (SMD=1.91, 95% CI: 1.74-4.07, P<0.0001) and teaching satisfaction (RR=1.38; 95% CI: 1.15 -1.65; P=0.01) compared with the LBL teaching model among the dental students. Additionally, a subgroup analysis showed significant differences in the effects of CBL on knowledge scores, skill scores and comprehensive ability scores when comparing theoretical and practical courses.

Conclusions The meta-analysis and systematic review of the eligible literature showed that the CBL pedagogy in Chinese dental education is an effective way to increase knowledge scores, skill scores, comprehensive ability scores, and teaching satisfaction. Successful adaptation of this teaching model could solve the scarcity of highly skilled and professional dentists in dental schools and hospitals across China.

Strengths and limitations of this study:

1. To the best of our knowledge, this is the most comprehensive network meta-analysis which describes case-based learning pedagogy compared to those of traditional Chinese lecture-based learning (LBL) teaching model.
2. This is the first study that provides a systematic estimate of the effectiveness of case-based learning in Chinese dental education.

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3. The source and quality of the articles included in the study represent a major limitation of our study.

Introduction

Dental education typically exploits the conventional methods of didactic, discipline-based curricula to increase students' competency with those skills and professional values in a subject-centered matter.[1] And the traditional didactic lecture-based curriculum uses an instructor-centered classroom setting in which students are passive listeners rather than active learners. Hence, the effectiveness of such teaching methods in equipping the graduates with a broad range of essential dental competencies, such as critical thinking and treating the patients comprehensively, is questionable. Notably, surveys in faculty members have reported that in most cases, graduate students could not efficiently implement their didactic pre-clinical knowledge into the actual clinical application when it comes to patient care. [2] As a result, long-established case-based learning (CBL) pedagogy, which originated from Harvard Medical School in the 1920s, has emerged as an effective teaching model innovation in health professional educations, including dental education.

CBL methods generally focus on using wide ranges of medical and clinical cases to teach students about actual patient care situations by guiding students on how to implement their acquired knowledge foundation in making decisions on problems that they may encounter in practice. In the CBL method, the student-teacher relationship goes on like students take on the learning responsibilities by analyzing the given case, while the teacher's role is a prompter and guide helping students justify their analyses in designing effective treatment strategy. The implementation process of CBL can be summarized as follows: establishment of the case → case grouping and analysis → brainstorming → identifying instruction goals → new findings → exchanging results between groups → reaching learning consensus and integrating the consensus into clinical practice.[3] CBL is increasingly becoming a popular teaching model applied in dental education in Europe, the United States, Canada, Australia and New Zealand since the 1990s, and some studies in these regions suggested that CBL is more effective in fostering deep learning, and efficient retention of analytical and diagnostic knowledge among students.[4] CBL may also facilitate competency development in students in a manner that parallels their future professional career goals.[5]

In the early 21st century, the Chinese dental education system realized the crucial systemic obstructions in transforming a dental graduate into a skilled and professional dentist who was in great demand during that time. To overcome this roadblock, some of the well-known key Chinese dental

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universities, including Shanghai Second Medical University, School of Stomatology at the Fourth Military Medical University, and School of Stomatology at Wuhan University, took critical steps forward to greatly reform the contemporary dental education model by including the most effective teaching concepts which were already being applied in diverse educational strategies, like CBL pedagogy. For the past 20 years, the CBL curriculum has been gradually adopted by several dental schools across China, subsequently increasing their success rates. In general, CBL curricula cover nearly all dental disciplines, including oral medicine, periodontology, prosthodontics, pediatric dentistry, orthodontics, preclinical training, etc. This kind of broad application of CBL methodology in the Chinese dental education system has been vividly reflected in the increasing number of publications of medical studies focusing on the CBL teaching model in China. Thus, this meta-analysis aimed to systematically review the effectiveness of CBL pedagogy performed in the Chinese dental education system. To evaluate the overall effectiveness, we performed subgroup analyses of the studies conducted in China to identify the benefits of the application of different types of courses in terms of knowledge scores, skill scores, and comprehensive ability scores, as well as teaching satisfaction.

Methods

Study Design

The study methods used for this meta-analysis and systematic review were based on the recommended PRISMA checklist guidelines.[6] Prior ethical approval was not necessary because it is a review of previously published articles and does not involve any processing of individual patient data.

Search Strategy

The published studies and trials were searched in the following Chinese electronic databases: China National Knowledge Infrastructure (CHKI), China Science Periodical Database (VIP), Chinese Biomedical documental database (CBM), WanFang Data. Also, the following English-language electronic databases: PubMed, EMBASE, and Cochrane Central Register of Controlled Trials, were searched. Next, the abstracts of the additional eligible studies posted in the international and national conference proceedings were also screened. The end date for all the electronic database searches was restricted to December 2019. The specific set of MeSH-terms or keywords used to search the above databases was like ("case-based learning" OR "CBL") AND ("dental* OR dental* education OR

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dental* students") AND ("China"). There was no search restriction on language. Furthermore, the reference lists of these selected articles were reviewed for additional related reports.

Selection criteria

There were five inclusion criteria: (1) the studies were randomized control trials (RCTs); (2) sufficient sample size was included in each RCTs; (3) the participants for the studies were undergraduate and graduate students of dental institutions in China; (4) the interventions included the CBL pedagogy in the experimental group covering all dental disciplines and traditional teaching methods (lecture-based learning, LBL) in the control group. Neither the experiment nor the control group should be exposed to supplementary teaching methods that could have an impact on the results; (5) the outcomes presented as data or descriptions of each RCT included at least one of the following scores: knowledge score, skill score, comprehensive ability score, pass rate, and teaching satisfaction.

Any study which did not meet the above criteria was excluded. The titles and abstracts of all retrieved articles were independently reviewed by two reviewers (Dong and Guo) to identify potential studies. The full texts of the eligible studies were then obtained and assessed independently by the two reviewers. Any discrepancies were resolved by discussion with a third reviewer (Lian Zhou) until a consensus was met.

Data extraction

Data were collected and extracted by two independent reviewers (Dong and Guo) using a predefined data form, which included studies regarding basic characteristics, such as the first authors, year of publication, involved dental disciplines, characteristics and numbers of the participants, intervention methods and outcome measures. Any conflict on the eligibility of the extracted information by these two reviewers was resolved by discussion among themselves or intervention of the third reviewer.

Outcomes

The primary outcomes in this meta-analysis were knowledge scores, skill scores, comprehensive ability scores in the CBL group and the LBL group. The secondary outcomes were pass rate and teaching satisfaction. Subgroup analyses were conducted to assess the benefits of applying CBL versus LBL to theoretical and practical courses in the above outcomes.

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Quality assessment

The methodological qualities of the included studies were assessed as adequate, inadequate or uncertain independently by the two reviewers (Dong and Guo) using guidelines described in the Cochrane Collaboration for Systematic Reviews of Interventions. [7] Any disagreement was resolved through discussion. The quality assessment was based on six general sources of bias described as random sequence generation, allocation concealment, blinding, incomplete outcome data, selective reporting, and other sources of bias contained in this specific study as the meaningful evaluation index. The overall methodological quality of each study was included as "low risk of bias", "high risk of bias", and "uncertain risk of bias". Information addressed by these items were retrieved from the published studies, and authors were contacted in case any additional information was needed.

Statistical analysis

Statistical analysis to test the heterogeneity of data and the meta-analysis were carried out using the Review Manager (RevMan) software (Version 5.3, the Cochrane Collaboration, UK). For continuous outcomes, such as knowledge scores, skill scores, comprehensive ability scores, the standardized mean difference (SMD) for effect size based on sample size and 95% confidence intervals (CI) were calculated from the same scale. For dichotomous outcomes, such as pass rate and teaching satisfaction, the risk ratio (RR) and 95% CI were calculated. The chi-squared test and I^2 statistic were used to assess statistical heterogeneity of the data. If the chi-squared test >0.10 or if the I^2 <50%, the fixed-effects model was chosen to pool data if there was no heterogeneity. Otherwise, a random-effects model was adopted. If the data could not be pooled, descriptive statistics was used only. Publication bias was tested independently using funnel plots of knowledge scores, skill scores, comprehensive ability scores, and teaching satisfaction; if the funnel plot was symmetric, there was a low potential for publication bias, and vice-versa.

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Patient and Public Involvement

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Since this is a systematic review and meta-analysis, patients and the public were not involved in the design, or planning of the study.

Results

Search results

The details on study inclusion and exclusion are summarized in the flow diagram (**Fig. 1**). The literature search in the databases yielded a total of 383 studies in the initial search, out of which 167 records were removed as duplicates, resulting in 216 records. After applying the selection criteria, 175 records were selected. Based on their titles and abstracts, a further 28 studies were excluded because some were experience summaries, and others were questionnaire surveys without theoretical measurement of the scores. The full texts of the remaining 147 studies were reviewed, and the additional 117 records were also excluded since they were trials without controls or CBL combined with supplementary teaching methods. Thus, a total of 30 studies [8-37], involving pooled 2356 dental graduate students, met the inclusion criteria, which were subsequently selected for this meta-analysis. All of the included studies were published between 2008 and 2019. These studies included 1198 students in the CBL teaching group and 1158 students in the LBL group. In these cohorts, most of the participating students were seniors. The sample size in the including trials ranged from 20 to 200 participants. Among these 30 studies, one study[37] was published in English, and the others were published in Chinese. And the types of dental disciplines included in the studies were prosthodontics[8,11,19, 26,27,28,33], endodontics[10,25,30,34], periodontology[18,21,31,35], orthodontics[12,20,32], oral medicine[37], general dentistry[13,17, 22, 27], oral and maxillofacial esthetic surgery[9, 36], oral anatomy and physiology[14], oral histology and pathology[15,16]. There were 18 RCTs[8,9,12,13,14,15,16, 17,18,19,20,26,27,28,32,33,35,37] on theoretical courses, and 12 RCTs[10,11, 21,22,23, 24,25, 29,30,31,34,36] on practical courses. The basic characteristics of the 30 included studies are presented in **Table 1**.

The quality of the 30 studies was evaluated using the Cochrane Collaboration Handbook for Systematic Reviews. There were 8 studies [13,20,26,28,31,34,35,37] reporting the detailed information of the randomization sequence generation, all of which were generated using the randomized number chart. However, none of the studies reported the allocation concealment methods and blinding. The summary of the methodological quality of each study is shown in (**Fig. 2**). The meta-analysis independently used funnel plots of knowledge scores, skill scores, comprehensive ability scores and teaching satisfaction to assess publication bias (**Fig.3 A, B, C& D**). The shape of the funnel plot was nearly symmetrical, indicating negligible evidence of significant publication bias.

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Measurements of Knowledge scores

Twenty-six of the included studies reported relevant data regarding knowledge scores. Among these, 4 studies [11,16,17,28] reported dichotomous data, such as pass rate, excellence rate, and improvement rate. Because of the diversity-mediated heterogeneity of data, the outcomes were not pooled. Remaining 22 studies [8,9,10,11-22, 24,25,27,28,30,31,32,36,37] reported the continuous data, such as mean scores and standard deviation, which were pooled in this meta-analysis (904 and 857 students in the CBL and LBL groups, respectively). The meta-analysis of the knowledge scores found that the CBL pedagogy significantly increased the knowledge scores by a mean of 1.53 compared with that of LBL pedagogy (95% CI: 0.93-2.53; P < 0.00001). The random-effects model was used for the meta-analysis because of the higher heterogeneity of the data (P < 0.00001, I^2 = 97%) (**Fig. 4**).

Measurements of Skill scores

8 studies [10,21,23,29,31,34,35,36] documented relevant data regarding skill scores (252 and 249 students in the CBL and LBL groups, respectively). The meta-analysis of the skill scores found that the CBL pedagogy significantly increased skill scores by a mean of 1.22 compared with that of LBL teaching model (95% CI: 0.34-2.11; P<0.0001). The random-effects model was used for the meta-analysis because of the higher heterogeneity of the data (P<0.00001, I^2=95%) (**Fig. 5**).

Measurements of Comprehensive ability scores

11 studies [12,13,20,21,22,24,25,26,29,31,32,30] produced relevant data regarding comprehensive ability scores (441 and 439 students in the CBL and LBL groups, respectively). The meta-analysis of the comprehensive ability scores found that the CBL teaching model significantly increased comprehensive ability scores by a mean of 2.91 compared with that of the LBL teaching model (95% CI, 1.74– 4.07; P<0.0001). The random-effects model was used for the meta-analysis because of the higher heterogeneity of the data (P<0.0001, I^2 = 98%) (**Fig. 6**).

Measurements of Teaching satisfaction

Teaching satisfaction was divided into 3 dimensions: satisfactory teaching effect, general teaching effect and poor teaching effect. A total of 8 studies [10,13, 15,16,17,27,28,30,36] reported relevant data regarding satisfactory teaching effects (340 and 340 students in the CBL and LBL groups,

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respectively). The meta-analysis of satisfactory teaching effect on the CBL teaching model found higher satisfactory rates compared with that of the LBL teaching model (RR: 1.38; 95% CI: 1.15–1.65; P=0.01). The random-effects model was used for the meta-analysis because of the higher heterogeneity of the data (P=.01, I^2=68%) (**Fig. 7**). There was no complete data to pool the general teaching effect and poor teaching effect between the CBL and LBL groups.

Subgroup analysis

Subgroup analysis was performed to examine the differences between practical and theoretical courses when CBL was applied instead of LBL methods. **Table 2** shows the results of the subgroup analysis. The outcome revealed that there were significant differences in the effects of CBL on knowledge scores, skill scores, and comprehensive ability scores when comparing theoretical and practical courses. In addition, a subgroup analysis of the level of training showed that the CBL teaching model also significantly improved the knowledge scores of senior and fifth year students **Table 3**.

Discussion

Since the beginning of the 21st century, worldwide medical education, including dentistry, has increasingly emphasized the development of realistic clinical competencies as well as educational milestones of the graduates. The definition of competency in dentistry has been made by the working party of the Competencies for Dental Licensure in Canada, as "competency is most often used to describe the skills, understanding and professional values of an individual ready to begin independent dental or allied oral health care practice".[38] At the professional level, the Commission on Dental Accreditation (CODA) proposed that dental students' competencies should include critical thinking and acquisition of evidence-based information in clinical reasoning and problem-solving.[39] CBL pedagogy, as an important element of competency-based education, typically focuses on the discussion of a patient case, and the fundamental concepts required for understanding the case are presented via lectures and/or readings before conducting case discussion. In these courses, teachers become a critical guide in assisting students in enriching their dynamic thought processes rather than passively acquiring information. CBL has been emerging as an outstanding start to transit students from more traditional and theoretical "text-book" oriented methodologies to practical scenarios. This unique academic concept enables students to think holistically, starting from presenting the medical history and diagnostics to the social aspects of the patient's quality of life during and after the course of treatments. This teaching strategy has shown to increase the students' interest to learn deeply about the realistic clinical practice and also prepares the students with a mindset to work in an

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interdisciplinary group, the advantages of which can then be seen to amplify further replenishing the deficiency of highly skilled dentists.

Unlike developed countries, CBL has not been a routine pedagogy in Chinese dental education. There have been certain inhibitory factors against the broad-spectrum implementation of CBL in China. Firstly, the traditional teacher-centered model has been dominating the dental education system in China for decades, which has almost established pre-defined educational experiences for both the students and teachers. Hence, it is evident that there will be a long-term adaption period to adjust the CBL teaching pattern and assessment system for both the teachers and students. Secondly, the levels and standards of dental education vary dramatically among different schools in China. The scarcity of high-quality and energetic teaching teams is a significant inhibitory factor against the successful implementation of the CBL pedagogy. Thirdly, the CBL curricula should be based on the sound fundamental knowledge background of the students. Notably, dental students in China usually have to complete the multi-disciplinary dental courses including restorative dentistry, preventive dentistry, periodontology, prosthodontics, pediatric dentistry, and oral medicine within 1-1.5 years of course duration, which is totally different from the dental education patterns in developed countries, such as the United States.[40] The time limitations make dental schools employ less time-consuming teaching models to fit well within the scheduled period. Hence, a great deal of educational reformation, re-arrangement, and successful integration between academia and professional hospitals are urgently required to vividly improve the Chinese dental education system.

The key findings of this meta-analysis suggested that the CBL pedagogy could significantly increase knowledge scores, skill scores, comprehensive ability scores and teaching satisfaction compared to that of the traditional LBL teaching model. Subgroup analyses also consistently demonstrated that the knowledge scores and comprehensive ability scores between practical and theoretical courses were significantly different, and satisfactory teaching effects in practical courses were significant. According to the questionnaire-based surveys of the students, CBL aroused their learning enthusiasm, trained them to improve their critical thinking ability, brought their interests toward working as a team, and enabled them to integrate and manage learning resources effectively and actively, and promoted self-study capacity.

However, there were also several limitations in the present study. The methodology qualities of the included RCTs were not significantly high. None of them reported the allocation concealment and blinding sources. There was significant heterogeneity of the data among the 22 pooled studies with respect to knowledge scores, the 8 pooled studies with respect to skill scores, and the 11 pooled

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studies with respect to comprehensive ability scores. The following factors might have led to these diversities: (1) the pooled studies focused on different dental disciplines; (2) the heterogeneity of the reported results, including knowledge scores and skill scores, was significantly high. (3) there were no guidelines to implicate CBL for different disciplines; (4) there were no standard criteria for the evaluation of the teaching effect of CBL pedagogy. Therefore, scientifically designed more random control trials in this field are needed in the future to address these issues.

To better apply and study the CBL teaching model in Chinese dental education background, firstly, teachers should design an effective teaching program based on teaching experiences that could facilitate and improve the students' ability to think critically and treat patients comprehensively on the basis of their sound fundamental knowledge of the dental sciences; secondly, teachers should develop or select realistic intriguing cases which link theoretical knowledge to practical aspects through inquiry-based learning methods, and simulate scenarios by simultaneously taking into account the patient's medical, dental, psychological, socio-economic and cultural conditions; thirdly, the teaching method of CBL is a three-dimensional cooperative relationship, which includes the dominant relationships between teachers and students, among students, and between virtual and realistic patient-doctor scenario. In the CBL teaching model, students are placed at the center of the educational teaching network. Students should be engaged in multi-modal activities such as information-gathering, higher-order analytical thinking, synthesis, and evaluation of the case-specific treatment strategy. In this way, the CBL curriculum would stimulate students' active learning capacity, and help them demonstrate the equivalent or superior preparation skills for key competencies compared to the requirements in their day-to-day professional life. Future studies should aim to assess the impact of the CBL on students' outcomes and to develop a better understanding of their learning experiences within the CBL framework in comparison to those they have during didactic courses.

In summary, this meta-analysis revealed that the CBL pedagogy could be an effective method for increasing the knowledge scores, skill scores, comprehensive ability scores as well as toward improving teaching satisfaction among the dental graduates in China. CBL appears to be superior to traditional lecture-based teaching methods. Thus, the use of the CBL teaching model may be optimal for improving dental education in China. However, the heterogeneity of the data from the included studies and the relatively low qualities of statistically validated measures should be taken into account in future practice, suggesting an urgent necessity for high-quality studies to efficiently assess the effectiveness of CBL pedagogy in the Chinese dental education system.

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Figure Legends:

Figure 1: Flow diagram of study selection (preferred reporting items for systematic review and meta-analysis, PRISMA).

Figure 2: Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

Figure 3: Funnel plots for publication bias. (A) Publication bias of knowledge scores. (B) Publication bias of skill scores. (C) Publication bias of comprehensive ability scores. (D) Publication bias of satisfactory teaching effect.

Figure 4: Forest plot of knowledge scores for CBL compared with LBL.

Figure 5: Forest plot of skill scores for CBL compared with LBL.

Figure 6: Forest plot of comprehensive ability scores for CBL compared with LBL.

Figure 7: Forest plot of teaching satisfaction for CBL compared with LBL.

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Table 1: Characteristics of the included studies.

Author	Year	Study design	No. of CBL	No. of LBL	Grade	Course name	Course type	Outcome measures
Zuo et al.[8]	2011	RCT	43	43	Senior	Prosthodontics	Theory course	KS, TS
Xia et al.[9]	2012	RCT	56	56	Junior	Oral and maxillofacial esthetic surgery	Theory course	KS, TS
Dou et al.[10]	2012	RCT	16	15	Senior	Endodontics	Practice course	KS, SS, CAS, TS
Jiang [11]	2019	RCT	110	116	Senior	Prosthodontics	Practice course	KS
Shi et al.[12]	2015	RCT	37	37	Senior	Othodontics	Theory course	KS, SS, CAS, TS
Wang [13]	2018	RCT	100	100	Senior	General dentistry	Theory course	KS, SS, CAS, TS
Liu et al. [14]	2012	RCT	61	30	Junior	Oral anatomy and physiology	Theory course	KS
Wang et al. [15]	2011	RCT	70	70	Junior	Oral histology and pathology	Theory course	KS, TS
Liang et al. [16]	2008	RCT	49	50	Junior	Oral histology and pathology	Theory course	KS, TS
Wang et al.[17]	2015	RCT	30	30	Senior	General dentistry	Theory course	KS, TS
Lu H et al. [18]	2012	RCT	76	65	Senior	Periodontology	Theory course	KS, TS
Wu et al. [19]	2014	RCT	45	44	Senior	Prosthodontics	Theory course	KS, TS
Zhang et al.[20]	2016	RCT	42	42	Senior	Orthodontics	Theory course	KS, SS, CAS, TS
Lu et al. [21]	2012	RCT	72	69	Senior	Periodontology	Practice course	KS, SS, CAS, TS
Zuo et al. [22]	2015	RCT	15	15	The fifth-year student	General dentistry	Practice course	KS, SS, CAS, TS
Zhao et al. [23]	2016	RCT	39	39	The fifth-year student	General dentistry	Practice course	KS, SS
Wang et al. [24]	2019	RCT	15	15	Senior	Endodontics	Practice course	CAS

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Yang et al. [25]	2015	RCT	20	20	Senior	Prosthodontics	Theory course	KS, CAS
Chen [26]	2019	RCT	25	25	Senior	Prosthodontics	Theory course	KS, CAS, TS
Liao [27]	2018	RCT	10	10	Senior	Prosthodontics	Theory course	KS, TS
Tao et al. [28]	2011	RCT	30	30	The fifth-year student	General dentistry	Practice course	KS, TS
Liu S et al. [29]	2013	RCT	20	21	Senior	Endodontics	Practice course	KS, SS, CAS, TS
Hu et al. [30]	2019	RCT	22	21	Senior	Periodontology	Practice course	KS, TS
Huang et al. [31]	2014	RCT	30	30	Master	Orthodontics	Practice course	SS
Yu et al. [32]	2018	RCT	30	30	Senior	Orthodontics	Theory course	KS, CAS, TS
Wei et al. [33]	2015	RCT	40	40	Senior	Prosthodontics	Theory course	KS, CAS, TS
Sun et al. [34]	2019	RCT	15	15	Senior	Endodontics	Practice course	KS, CAS, TS
Yang et al. [35]	2018	RCT	20	20	Senior	Periodontology	Practice course	KS, SS, TS
Li BH et al. [36]	2015	RCT	40	40	Senior	Oral and maxillofacial esthetic surgery	Theory course	KS, SS, CAS, TS
Du et al. [37]	2013	RCT	20	20	Senior	Oral Medicine	Theory course	KS

CAS: Comprehensive ability scores; CBL: Case-based learning; KS: Knowledge scores; LBL: Lecture-based learning; No.: Number; RCT: Randomized controlled trial; SS: Skill scores; TS: Teaching satisfactions.

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Table 2: Results of subgroup analyses in the current meta-analysis.

	No. of participants			Incidence				
Items	**Studies**	**CBL**	**LBL**	***P***	**SMD/RR**	**95%CI**	**Heterogeneity(I^2)**	**Model**
Knowledge scores *								
Theoretical courses	14	656	651	<0.0001	1.78	0.95-2.62	97%	Random
Practical courses	7	226	185	<0.0001	1.1	0.34-1.86	91%	Random
Skill scores *								
Theoretical courses	2	40	40	<0.0001	4.06	3.28-4.84	N. S	Random
Practical courses	6	212	209	0.02	0.83	0.15-1.51	90%	Random
Comprehensive ability scores *								
Theoretical courses	7	309	309	<0.0001	3.81	1.88-5.74	98%	Random
Practical courses	4	132	130	0.01	1.39	0.82-1.97	72%	Random
Teaching satisfactions **								
Theoretical causes	6	293	294	<0.0001	2.63	1.78-3.91	65%	Random
Practical courses	2	47	46	0.0001	6.31	2.47-16.13	N. S	Random

CBL: Case-based learning; CI: Confidence interval; LBL: Lecture-based learning; N.S: Non-significant; No.: Number; RR: Risk ratio; SMD: Standard mean difference; * Values presented as mean difference and 95% confidence interval; * * Values presented as risk ratio and 95% confidence interval.

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Table 3: Subgroup analyses of the level of training in the current meta-analysis.

	No. of participant			Incidence				
Items	**Studies**	**CBL**	**LBL**	***P***	**SMD/MD**	**95%CI**	**Heterogeneity (I^2)**	**Model**
Knowledge scores *								
Junior	4	263	231	0.11	0.82	-0.17-1.82	96%	Random
Senior	22	686	671	0.001	1.53	0.75-2.31	97%	Random
The fifth year student	3	82	82	0.007	1.05	0.29-1.81	81%	Random

CBL: Case-based learning; CI: Confidence interval; LBL: Lecture-based learning; No.: Number; MD: Mean difference; SMD: Standard mean difference; * Values presented as mean difference and 95% confidence interval.

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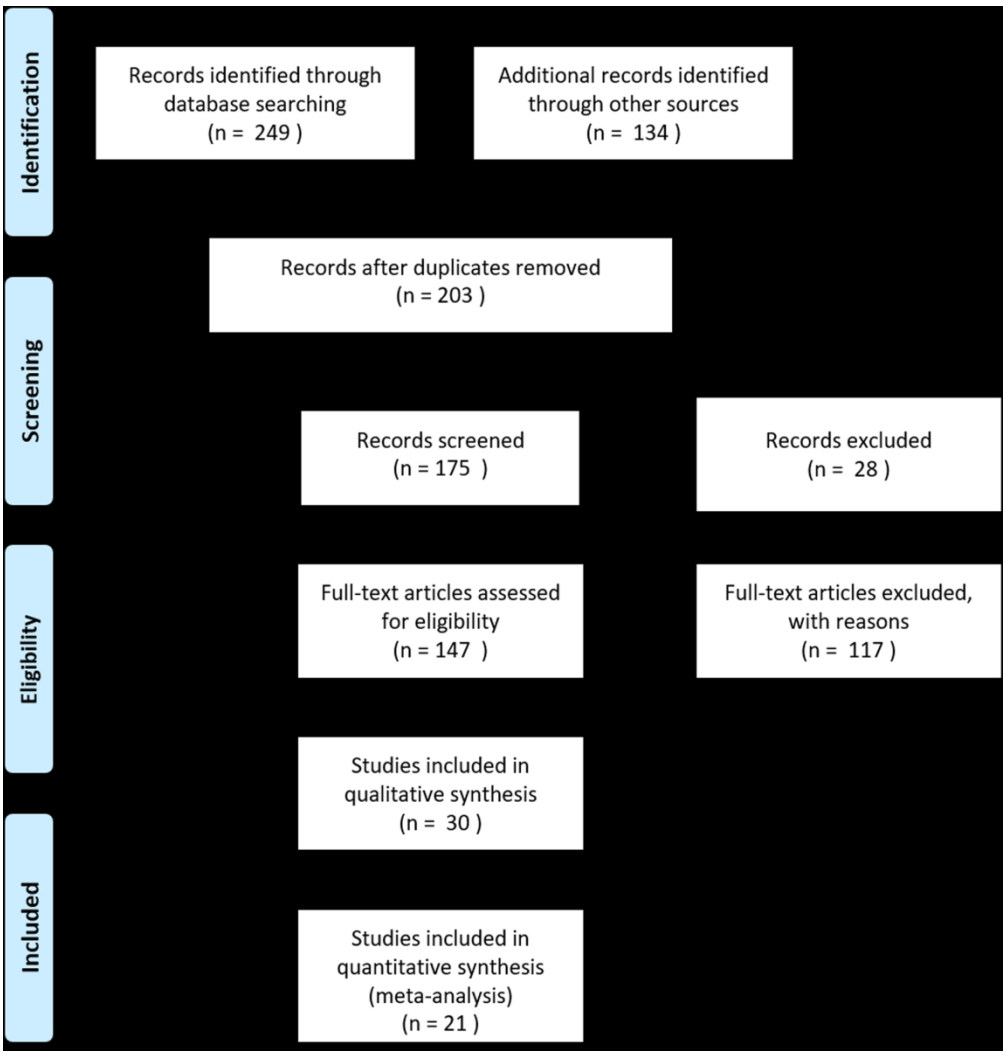


Figure 1: Flow diagram of study selection (preferred reporting items for systematic review and meta-analysis, PRISMA).

165x173mm (300 x 300 DPI)

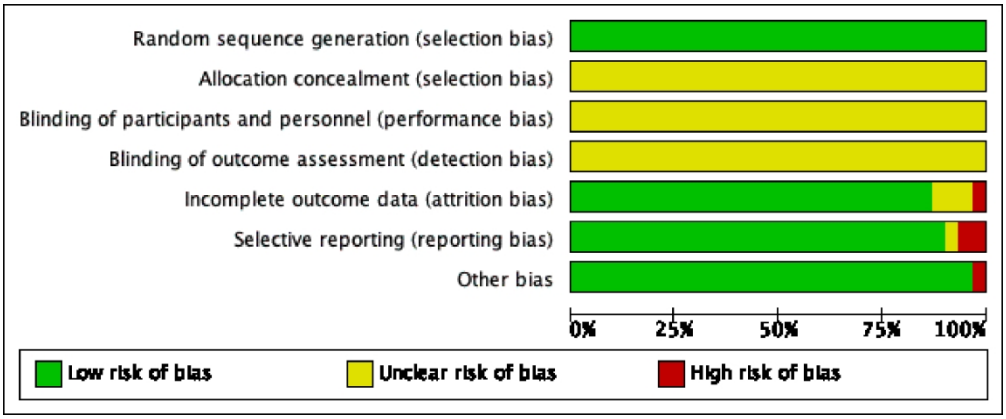


Figure 2: Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

219x90mm (300 x 300 DPI)

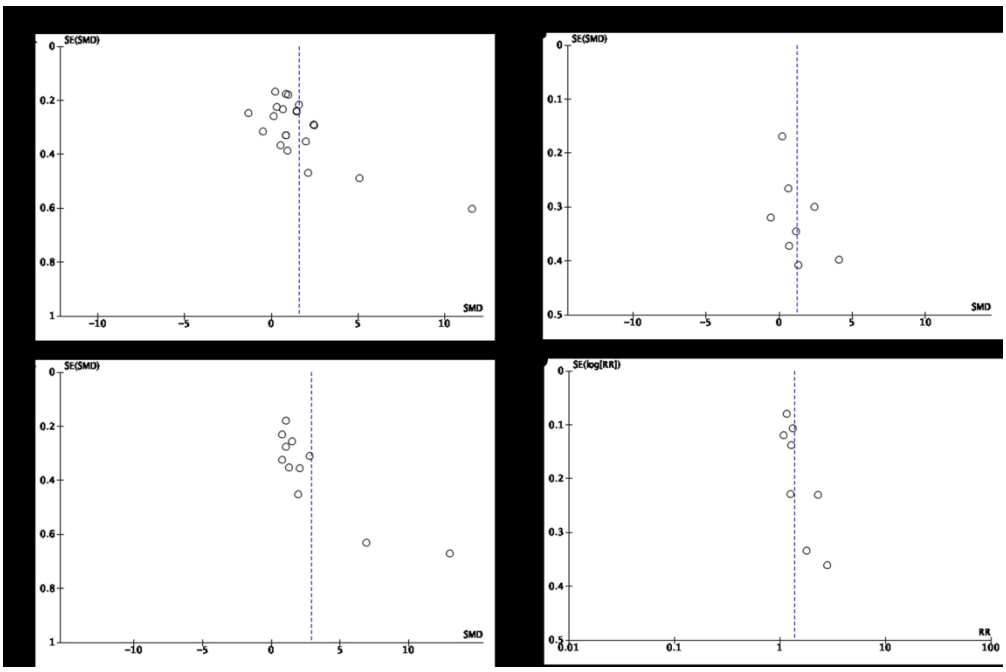


Figure 3: Funnel plots for publication bias. (A) Publication bias of knowledge scores. (B) Publication bias of skill scores. (C) Publication bias of comprehensive ability scores. (D) Publication bias of satisfactory teaching effect.

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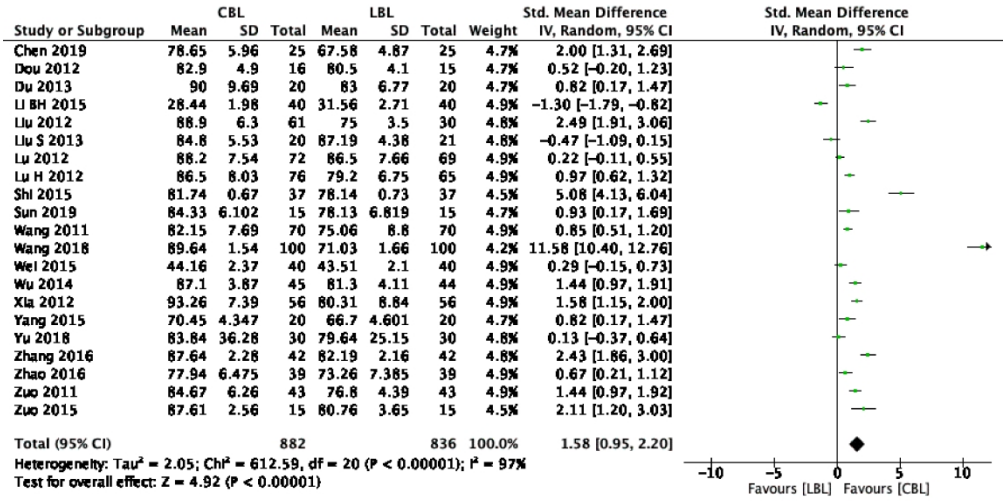


Figure 4: Forest plot of knowledge scores for CBL compared with LBL.

314x157mm (300 x 300 DPI)

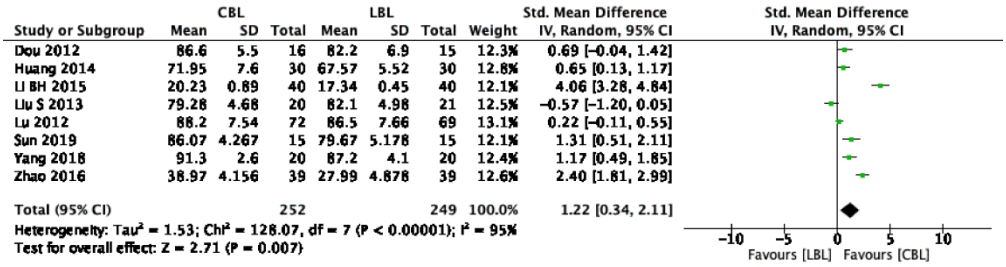


Figure 5: Forest plot of skill scores for CBL compared with LBL.

314x84mm (300 x 300 DPI)

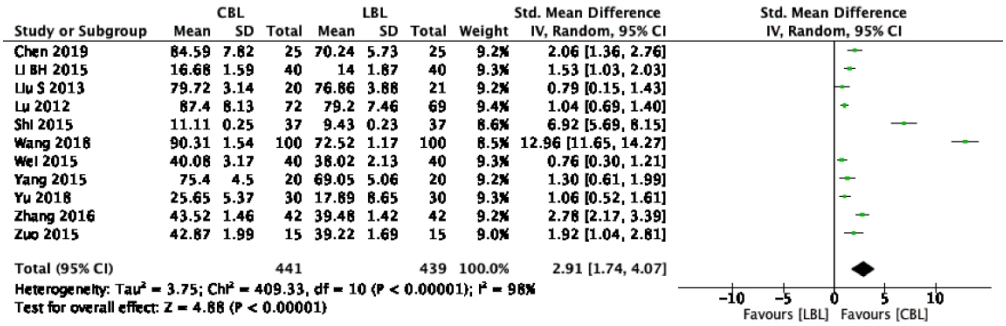


Figure 6: Forest plot of comprehensive ability scores for CBL compared with LBL.

308x101mm (300 x 300 DPI)

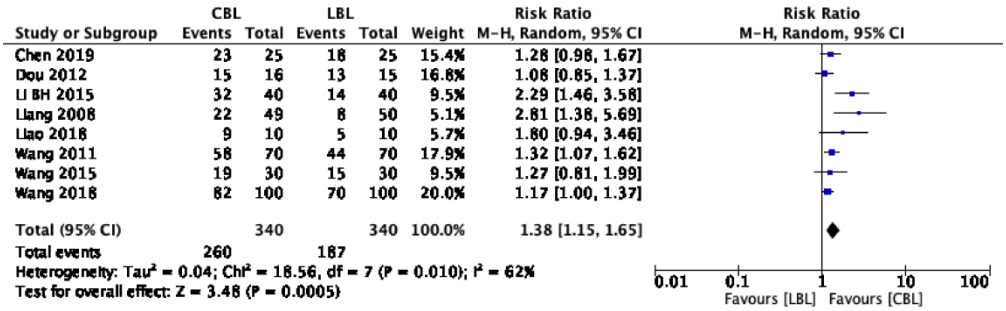


Figure 7: Forest plot of teaching satisfaction for CBL compared with LBL.

287x90mm (300 x 300 DPI)

PRISMA 2009 Checklist

Section/Topic	#	Checklist Item	Reported on Page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5-6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5-6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

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PRISMA 2009 Checklist

Section/Topic	#	Checklist Item	Reported on Page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9-10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-10
Synthesis of results	21	Present the main results of the review. If meta-analyses done, include for each, confidence intervals and measures of consistency.	8-10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8-10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	12
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	12
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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Effectiveness of Case-based Learning in Chinese Dental Education: A systematic review and meta-analysis

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Secondary Subject Heading:	Dentistry and oral medicine
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Title: Effectiveness of Case-based Learning in Chinese Dental Education: A systematic review and meta-analysis

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Contributorship Statement

Haitao Dong, Xinyuan Zhang and Xin Zhang carried out the entire procedure including the literature search, data extraction, performed the statistical analysis, drafted and revised the manuscript. **Jizhi Zhao and Xiaomin Wu** contributed to the revision of the manuscript. **Chunlan Guo and Lian Zhou** conceived of the study, coordinated and participated in the entire process of drafting and revising the manuscript.

Data Sharing Statement:

All data relevant to the study are included in the article or uploaded as online supplemental information. No unpublished data are available.

Ethical approval and informed consent

Since this is a systematic review, ethical approval and informed consent are not required.

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Abstract

Objective The application of case-based learning (CBL) pedagogy has been emerging as an improved and more practical learning method across the Chinese dental education system. This article provides a critical overview of CBL pedagogy and further assess the effectiveness of this teaching model in Chinese dental education.

Design A Systematic review and meta-analysis.

Methods Studies published up to December 2019 were searched in the following electronic databases: China National Knowledge Infrastructure (CNKI), China Science Periodical Database (VIP), Chinese Biomedical documental database (CBM), Wanfang data, PubMed, EMBASE and Cochrane Central Register of Control Trials. All randomized controlled trials (RCTs) that compared the effectiveness of the CBL teaching model and the traditional lecture-based learning (LBL) model in all dental disciplines were included. The assessment of methodological quality was based on the guidelines described in the Cochrane Handbook for Systematic Reviews, and the meta-analysis was performed using the software RevMan v5.3

Results A total of 30 RCTs were finally included, with a total sample size of 2356 dental students. The CBL pedagogy significantly increased knowledges scores (SMD=1.58, 95% CI: 0.95-2.20, P<0.0001), skill scores (SMD =1.22, 95% CI: 0.34-2.11, P<0.0001), comprehensive ability scores (SMD=1.91, 95% CI: 1.74-4.07, P<0.0001) and teaching satisfaction (RR=1.38; 95% CI: 1.15 -1.65; P=0.01) compared with the LBL teaching model among the dental students. Additionally, a subgroup analysis showed significant differences in the effects of CBL on knowledge scores, skill scores and comprehensive ability scores when comparing theoretical and practical courses.

Conclusions The meta-analysis and systematic review of the eligible literature showed that the CBL pedagogy in Chinese dental education is an effective way to increase knowledge scores, skill scores, comprehensive ability scores, and teaching satisfaction. Successful adaptation of this teaching model could solve the scarcity of highly skilled and professional dentists in dental schools and hospitals across China.

Strengths and limitations of this study:

. This is the first systematic review and meta-analysis of case-based learning (CBL) practice in Chinese dental education.

. It gives us some beneficial proposal to implementing the CBL pedagogy in future.

. The included studies were research in the field of medical education, so it is impossible for the researchers to implement allocation concealment and blind.

. Methodology qualities of the included studies represent a major limitation of this review.

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. Measurement bias exists in this review, for there is no standard criterion for evaluating the effectiveness of the CBL pedagogy in the included studies.

Introduction

Dental education typically exploits the conventional methods of didactic, discipline-based curricula to increase students' competency with those skills and professional values in a subject-centered matter.[1] The traditional didactic lecture-based curriculum uses an instructor-centered classroom setting in which students are passive listeners rather than active learners. Hence, the effectiveness of such teaching methods in equipping the graduates with a broad range of essential dental competencies, such as critical thinking and treating the patients comprehensively, is questionable. Notably, surveys in faculty members have reported that in most cases, graduate students could not efficiently implement their didactic pre-clinical knowledge into the actual clinical application when it comes to patient care. [2] As a result, long-established case-based learning (CBL) pedagogy, which originated from Harvard Medical School in the 1920s, has emerged as an effective teaching model innovation in health professional educations, including dental education.

CBL methods generally focus on using wide ranges of medical and clinical cases to teach students about actual patient care situations. In these curriculum, the teachers mainly guide the students to implement their acquired knowledge foundation in making decisions on problems that they may encounter in practice. So, the student-teacher relationship goes on like that, the students take on the learning responsibilities by analyzing the given case and designing effective treatment strategy, while the teacher's role is a prompter and guide to help students justify their analyses. The implementation process of CBL can be summarized as follows: establishment of the case → case grouping and analysis → brainstorming → identifying instruction goals → new findings → exchanging results between groups → reaching learning consensus and integrating the consensus into clinical practice.[3] CBL is increasingly becoming a popular teaching model applied in dental education in Europe, the United States, Canada, Australia and New Zealand since the 1990s, and some studies in these regions suggested that CBL is more effective in fostering deep learning, and efficient retention of analytical and diagnostic knowledge among students.[4] CBL may also facilitate competency development in students in a manner that parallels their future professional career goals.[5]

In the early 21st century, the Chinese dental education system realized the crucial systemic obstructions in transforming a dental graduate into a skilled and professional dentist who was in great demand. To overcome this roadblock, some of the well-known key Chinese dental schools, including

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Shanghai Second Medical University, School of Stomatology at the Fourth Military Medical University, and School of Stomatology at Wuhan University, took critical steps forward to greatly reform the contemporary dental education models. These models included the most effective teaching concepts and were already being applied in diverse educational strategies, like CBL pedagogy. For the past 20 years, the CBL curriculum has been gradually adopted by several more dental schools across China, subsequently presenting its increasing success rates. In general, CBL curricula cover nearly all dental disciplines, including oral medicine, periodontology, prosthodontics, pediatric dentistry, orthodontics, preclinical training, etc. This kind of broad application of CBL methodology in the Chinese dental education system has been vividly reflected in the increasing number of publications of medical studies focusing on the CBL teaching model in China. Thus, this meta-analysis aimed to systematically review the effectiveness of CBL pedagogy performed in the Chinese dental education system. To evaluate the overall effectiveness, we performed subgroup analyses of the studies conducted in China to identify the benefits of the application of different types of courses in terms of knowledge scores, skill scores, and comprehensive ability scores, as well as teaching satisfaction.

Methods

Study Design

The study methods used for this meta-analysis and systematic review were based on the recommended PRISMA checklist guidelines.[6] Prior ethical approval was not necessary because it is a review of previously published articles and does not involve any processing of individual patient data.

Search Strategy

The published studies and trials were searched in the following Chinese electronic databases: China National Knowledge Infrastructure (CHKI), China Science Periodical Database (VIP), Chinese Biomedical documental database (CBM), WanFang Data. Also, the following English-language electronic databases: PubMed, EMBASE, and Cochrane Central Register of Controlled Trials, were searched. Next, the abstracts of the additional eligible studies posted in the international and national conference proceedings were also screened. The end date for all the electronic database searches was restricted to December 2019. The specific set of MeSH-terms or keywords used to search the above databases was like ("case-based learning" OR "CBL") AND ("dental* OR dental* education OR

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dental* students") AND ("China"). There was no search restriction on language. Furthermore, the reference lists of these selected articles were reviewed for additional related reports.

Selection criteria

There were four inclusion criteria: (1) the studies were randomized control trials (RCTs); (2) the participants for the studies were undergraduate and graduate students of dental institutions in China; (3) the interventions included the CBL pedagogy in the experimental group covering all dental disciplines and traditional teaching methods (lecture-based learning, LBL) in the control group. Neither the experiment nor the control group should be exposed to supplementary teaching methods that could have an impact on the results; (4) the outcomes presented as data or descriptions of each RCT included at least one of the following scores: knowledge score, skill score, comprehensive ability score, pass rate, and teaching satisfaction.

Any study which did not meet the above criteria was excluded. The titles and abstracts of all retrieved articles were independently reviewed by two reviewers (Dong and Guo) to identify potential studies. The full texts of the eligible studies were then obtained and assessed independently by the two reviewers. Any discrepancies were resolved by discussion with a third reviewer (Lian Zhou) until a consensus was met.

Data extraction

Data were collected and extracted by two independent reviewers (Dong and Guo) using a predefined data form, which included studies regarding basic characteristics, such as the first authors, year of publication, involved dental disciplines, characteristics and numbers of the participants, intervention methods and outcome measures. Any conflict on the eligibility of the extracted information by these two reviewers was resolved by discussion among themselves or intervention of the third reviewer.

Outcomes

The primary outcomes in this meta-analysis were knowledge scores, skill scores, comprehensive ability scores in the CBL group and the LBL group. The secondary outcomes were pass rate and teaching satisfaction. In the including studies, the teaching satisfaction was usually evaluated by Likert scale. From 5 to 1, the decrease of the scores mean the decline of satisfaction. When the scores were 5~4, teaching satisfaction was defined as significant. When the scores were 3~2, teaching

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satisfaction was defined as general. When the scores were 1, teaching satisfaction was defined as poor. Subgroup analyses were conducted to assess the benefits of applying CBL versus LBL to theoretical and practical courses in the above outcomes.

Quality assessment

The methodological qualities of the included studies were assessed as adequate, inadequate or uncertain independently by the two reviewers (Dong and Guo) using guidelines described in the Cochrane Collaboration for Systematic Reviews of Interventions. [7] Any disagreement was resolved through discussion. The quality assessment was based on six general sources of bias described as random sequence generation, allocation concealment, blinding, incomplete outcome data, selective reporting, and other sources of bias contained in this specific study as the meaningful evaluation index. The overall methodological quality of each study was included as "low risk of bias", "high risk of bias", and "uncertain risk of bias". Information addressed by these items were retrieved from the published studies, and authors were contacted in case any additional information was needed.

Statistical analysis

Statistical analysis to test the heterogeneity of data and the meta-analysis were carried out using the Review Manager (RevMan) software (Version 5.3, the Cochrane Collaboration, UK). For continuous outcomes, such as knowledge scores, skill scores, comprehensive ability scores, the standardized mean difference (SMD) for effect size based on sample size and 95% confidence intervals (CI) were calculated from the same scale. For dichotomous outcomes, such as pass rate and teaching satisfaction, the risk ratio (RR) and 95% CI were calculated. If the data could not be pooled, descriptive statistics was used only. Publication bias was tested independently using funnel plots of knowledge scores, skill scores, comprehensive ability scores, and teaching satisfaction; if the funnel plot was symmetric, there was a low potential for publication bias, and vice-versa.

Role of the funding source

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Patient and Public Involvement

Since this is a systematic review and meta-analysis, patients and the public were not involved in the design, or planning of the study.

Results

Search results

The details on study inclusion and exclusion are summarized in the flow diagram (**Fig. 1**). The literature search in the databases yielded a total of 383 studies in the initial search, out of which 167 records were removed as duplicates, resulting in 216 records. After applying the selection criteria, 175 records were selected. Based on their titles and abstracts, a further 28 studies were excluded because some were experience summaries, and others were questionnaire surveys without theoretical measurement of the scores. The full texts of the remaining 147 studies were reviewed, and the additional 117 records were also excluded since they were trials without controls or CBL combined with supplementary teaching methods. Thus, a total of 30 studies [8-37], involving pooled 2356 dental graduate students, met the inclusion criteria, which were subsequently selected for this meta-analysis. All of the included studies were published between 2008 and 2019. These studies included 1198 students in the CBL teaching group and 1158 students in the LBL group. In these cohorts, most of the participating students were seniors. The sample size in the including trials ranged from 20 to 200 participants. Among these 30 studies, one study [37] was published in English, and the others were published in Chinese. And the types of dental disciplines included in the studies were prosthodontics [8,11,19,26,27,28,33], endodontics [10,25,30,34], periodontology [18,21,31,35], orthodontics [12,20,32], oral medicine [37], general dentistry [13,17,22,27], oral and maxillofacial esthetic surgery [9, 36], oral anatomy and physiology [14], oral histology and pathology [15,16]. There were 18 RCTs [8,9,12,13,14,15,16, 17,18,19,20,26,27,28,32,33,35,37] on theoretical courses, and 12 RCTs [10,11,21,22,23, 24,25,29,30,31,34,36] on practical courses. The basic characteristics of the 30 included studies are presented in **Table 1**.

The quality of the 30 studies was evaluated using the Cochrane Collaboration Handbook for Systematic Reviews. There were 8 studies [13,20,26,28,31,34,35,37] reporting the detailed information of the randomization sequence generation, all of which were generated using the randomized number chart. However, none of the studies reported the allocation concealment methods and blinding. The summary of the methodological quality of each study is shown in (**Fig. 2**). The meta-analysis independently used funnel plots of knowledge scores, skill scores, comprehensive ability scores and

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teaching satisfaction to assess publication bias (**Fig.3 A, B, C& D**). The shape of the funnel plot was nearly symmetrical, indicating negligible evidence of significant publication bias.

Measurements of Knowledge scores

Twenty-six of the included studies reported relevant data regarding knowledge scores. Among these, 4 studies [11,16,17,28] reported dichotomous data, such as pass rate, excellence rate, and improvement rate. Because of the diversity-mediated heterogeneity of data, the outcomes were not pooled. Remaining 22 studies [8,9,10,11-22,24,25,27,28,3031,32,36,37] reported the continuous data, such as mean scores and standard deviation, which were pooled in this meta-analysis (904 and 857 students in the CBL and LBL groups, respectively). The meta-analysis of the knowledge scores found that the CBL pedagogy significantly increased the knowledge scores by a mean of 1.58 compared with that of LBL pedagogy (95% CI: 0.95-2.20; P < 0.00001). The random-effects model was used for the meta-analysis because of the higher heterogeneity of the data (P < 0.00001, I^2= 97%) (**Fig. 4**).

Measurements of Skill scores

Eight studies [10,21,23,29,31,34,35,36] documented relevant data regarding skill scores (252 and 249 students in the CBL and LBL groups, respectively). The meta-analysis of the skill scores found that the CBL pedagogy significantly increased skill scores by a mean of 1.22 compared with that of LBL teaching model (95% CI: 0.34-2.11; P<0.0001). The random-effects model was used for the meta-analysis because of the higher heterogeneity of the data (P<0.00001, I^2=95%) (**Fig. 5**).

Measurements of Comprehensive ability scores

Eleven studies [12,13,20,21,22,24,25,26,29,31,32,30] produced relevant data regarding comprehensive ability scores (441 and 439 students in the CBL and LBL groups, respectively). The meta-analysis of the comprehensive ability scores found that the CBL teaching model significantly increased comprehensive ability scores by a mean of 2.91 compared with that of the LBL teaching model (95% CI, 1.74– 4.07; P<0.0001). The random-effects model was used for the meta-analysis because of the higher heterogeneity of the data (P<0.0001, I^2= 98%) (**Fig. 6**).

Measurements of Teaching satisfaction

Teaching satisfaction was divided into 3 dimensions: satisfactory teaching effect, general teaching effect and poor teaching effect. A total of 8 studies [10,13,15,16,17,28,30,36] reported relevant data regarding

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satisfactory teaching effects (340 and 340 students in the CBL and LBL groups, respectively). The meta-analysis of satisfactory teaching effect on the CBL teaching model found higher satisfactory rates compared with that of the LBL teaching model (RR: 1.38; 95% CI: 1.15–1.65; P=0.01). The random-effects model was used for the meta-analysis because of the higher heterogeneity of the data (P=.01, I^2=68%) (**Fig. 7**). There was no complete data to pool the general teaching effect and poor teaching effect between the CBL and LBL groups.

Subgroup analysis

Subgroup analysis was performed to examine the differences between practical and theoretical courses when CBL was applied instead of LBL methods. **Table 2** shows the results of the subgroup analysis. The outcome revealed that there were significant differences in the effects of CBL on knowledge scores, skill scores, and comprehensive ability scores when comparing theoretical and practical courses. In addition, a subgroup analysis of the level of training showed that the CBL teaching model also significantly improved the knowledge scores of senior and fifth year students **Table 3**.

Discussion

Since the beginning of the 21st century, worldwide medical education, including dentistry, has increasingly emphasized the development of realistic clinical competencies as well as educational milestones of the graduates. The definition of competency in dentistry has been made by the working party of the Competencies for Dental Licensure in Canada, as "competency is most often used to describe the skills, understanding and professional values of an individual ready to begin independent dental or allied oral health care practice".[38] At the professional level, the Commission on Dental Accreditation (CODA) proposed that dental students' competencies should include critical thinking and acquisition of evidence-based information in clinical reasoning and problem-solving.[39] CBL pedagogy, as an important element of competency-based education, typically focuses on the discussion of a patient case, and the fundamental concepts required for understanding the case are presented via lectures and/or readings before conducting case discussion. In these courses, teachers become a critical guide in assisting students in enriching their dynamic thought processes rather than passively acquiring information. CBL has been emerging as an outstanding start to transit students from more traditional and theoretical "text-book" oriented methodologies to practical scenarios. This unique academic concept enables students to think holistically, starting from presenting the medical history and diagnostics to the social aspects of the patient's quality of life during and after the course of treatments. This teaching strategy has shown to increase the students' interest to learn deeply about the realistic clinical practice and also prepares the students with a mindset to work in an

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interdisciplinary group, the advantages of which can then be seen to amplify further replenishing the deficiency of highly skilled dentists.

Unlike developed countries, CBL has not been a routine pedagogy in Chinese dental education. There have been certain inhibitory factors against the broad-spectrum implementation of CBL in China. Firstly, the traditional teacher-centered model has been dominating the dental education system in China for decades, which has almost established pre-defined educational experiences for both the students and teachers. Hence, it is evident that there will be a long-term adaption period to adjust the CBL teaching pattern and assessment system for both the teachers and students. Secondly, the levels and standards of dental education vary dramatically among different schools in China. The scarcity of high-quality and energetic teaching teams is a significant inhibitory factor against the successful implementation of the CBL pedagogy. Thirdly, the CBL curricula should be based on the sound fundamental knowledge background of the students. Notably, dental students in China usually have to complete the multi-disciplinary dental courses including restorative dentistry, preventive dentistry, periodontology, prosthodontics, pediatric dentistry, and oral medicine within 1-1.5 years of course duration, which is totally different from the dental education patterns in developed countries, such as the United States.[40] The time limitations make dental schools employ less time-consuming teaching models to fit well within the scheduled period. Hence, a great deal of educational reformation, re-arrangement, and successful integration between academia and professional hospitals are urgently required to vividly improve the Chinese dental education system.

The key findings of this meta-analysis suggested that the CBL pedagogy could significantly increase knowledge scores, skill scores, comprehensive ability scores and teaching satisfaction compared to that of the traditional LBL teaching model. Subgroup analyses also consistently demonstrated that the knowledge scores and comprehensive ability scores between practical and theoretical courses were significantly different, and satisfactory teaching effects in practical courses were significant. According to the questionnaire-based surveys of the students, CBL aroused their learning enthusiasm, trained them to improve their critical thinking ability, brought their interests toward working as a team, and enabled them to integrate and manage learning resources effectively and actively, and promoted self-study capacity.

As we know, this is the first systematic review and meta-analysis of CBL practice in Chinese dental education. Compared with other reviews on teaching methods reform in Chinese dental education previously, such as meta-analysis on PBL practice in Chinese dental education published by Huang et al [40], the present study had two strengths. Firstly, a subgroup analysis was performed

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comparing the course type between the theoretical courses and the practical courses. Secondly, this study independently used funnel plots to assess publication bias. The plots were generally symmetrical and showed a lower publication bias.

However, there were also several limitations in the present study. The methodology qualities of the included RCTs were not significantly high. None of them reported the allocation concealment and blinding sources. There was significant heterogeneity of the data among the 22 pooled studies with respect to knowledge scores, the 8 pooled studies with respect to skill scores, and the 11 pooled studies with respect to comprehensive ability scores. The following factors might have led to these diversities: (1) the pooled studies focused on different dental disciplines; (2) the heterogeneity of the reported results, including knowledge scores and skill scores, was significantly high. (3) there were no guidelines to implicate CBL for different disciplines; (4) there were no standard criteria for the evaluation of the teaching effect of CBL pedagogy. Therefore, scientifically designed more random control trials in this field are needed in the future to address these issues.

To better apply and study the CBL teaching model in Chinese dental education background, firstly, teachers should design an effective teaching program based on teaching experiences that could facilitate and improve the students' ability to think critically and treat patients comprehensively on the basis of their sound fundamental knowledge of the dental sciences; secondly, teachers should develop or select realistic intriguing cases which link theoretical knowledge to practical aspects through inquiry-based learning methods, and simulate scenarios by simultaneously taking into account the patient's medical, dental, psychological, social-economic and cultural conditions; thirdly, the teaching method of CBL is a three-dimensional cooperative relationship, which includes the dominant relationships between teachers and students, among students, and between virtual and realistic patient-doctor scenario. In the CBL teaching model, students are placed at the center of the educational teaching network. Students should be engaged in multi-modal activities such as information-gathering, higher-order analytical thinking, synthesis, and evaluation of the case-specific treatment strategy. In this way, the CBL curriculum would stimulate students' active learning capacity, and help them demonstrate the equivalent or superior preparation skills for key competencies compared to the requirements in their day-to-day professional life. Future studies should aim to assess the impact of the CBL on students' outcomes and to develop a better understanding of their learning experiences within the CBL framework in comparison to those they have during didactic courses.

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In summary, this meta-analysis revealed that the CBL pedagogy could be an effective method for increasing the knowledge scores, skill scores, comprehensive ability scores as well as toward improving teaching satisfaction among the dental graduates in China. CBL appears to be superior to traditional lecture-based teaching methods. Thus, the use of the CBL teaching model may be optimal for improving dental education in China. However, the heterogeneity of the data from the included studies and the relatively low qualities of statistically validated measures should be taken into account in future practice, suggesting an urgent necessity for high-quality studies to efficiently assess the effectiveness of CBL pedagogy in the Chinese dental education system.

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Figure Legends:

Figure 1: Flow diagram of study selection (preferred reporting items for systematic review and meta-analysis, PRISMA).

Figure 2: Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

Figure 3: Funnel plots for publication bias. (A) Publication bias of knowledge scores. (B) Publication bias of skill scores. (C) Publication bias of comprehensive ability scores. (D) Publication bias of satisfactory teaching effect.

Figure 4: Forest plot of knowledge scores for CBL compared with LBL.

Figure 5: Forest plot of skill scores for CBL compared with LBL.

Figure 6: Forest plot of comprehensive ability scores for CBL compared with LBL.

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Figure 7: Forest plot of teaching satisfaction for CBL compared with LBL.

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Table 1: Characteristics of the included studies.

Author	Year	Study design	No. of CBL	No. of LBL	Grade	Course name	Course type	Outcome measures
Zuo et al.[8]	2011	RCT	43	43	Senior	Prosthodontics	Theory course	KS, TS
Xia et al.[9]	2012	RCT	56	56	Junior	Oral and maxillofacial esthetic surgery	Theory course	KS, TS
Dou et al.[10]	2012	RCT	16	15	Senior	Endodontics	Practice course	KS, SS, CAS, TS
Jiang [11]	2019	RCT	110	116	Senior	Prosthodontics	Practice course	KS
Shi et al.[12]	2015	RCT	37	37	Senior	Othodontics	Theory course	KS, SS, CAS, TS
Wang [13]	2018	RCT	100	100	Senior	General dentistry	Theory course	KS, SS, CAS, TS
Liu et al. [14]	2012	RCT	61	30	Junior	Oral anatomy and physiology	Theory course	KS
Wang et al. [15]	2011	RCT	70	70	Junior	Oral histology and pathology	Theory course	KS, TS
Liang et al. [16]	2008	RCT	49	50	Junior	Oral histology and pathology	Theory course	KS, TS
Wang et al.[17]	2015	RCT	30	30	Senior	General dentistry	Theory course	KS, TS
Lu H et al. [18]	2012	RCT	76	65	Senior	Periodontology	Theory course	KS, TS
Wu et al. [19]	2014	RCT	45	44	Senior	Prosthodontics	Theory course	KS, TS
Zhang et al.[20]	2016	RCT	42	42	Senior	Orthodontics	Theory course	KS, SS, CAS, TS
Lu et al. [21]	2012	RCT	72	69	Senior	Periodontology	Practice course	KS, SS, CAS, TS
Zuo et al. [22]	2015	RCT	15	15	The fifth-year student	General dentistry	Practice course	KS, SS, CAS, TS
Zhao et al. [23]	2016	RCT	39	39	The fifth-year student	General dentistry	Practice course	KS, SS
Wang et al. [24]	2019	RCT	15	15	Senior	Endodontics	Practice course	CAS

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Yang et al. [25]	2015	RCT	20	20	Senior	Prosthodontics	Theory course	KS, CAS
Chen [26]	2019	RCT	25	25	Senior	Prosthodontics	Theory course	KS, CAS, TS
Liao [27]	2018	RCT	10	10	Senior	Prosthodontics	Theory course	KS, TS
Tao et al. [28]	2011	RCT	30	30	The fifth-year student	General dentistry	Practice course	KS, TS
Liu S et al. [29]	2013	RCT	20	21	Senior	Endodontics	Practice course	KS, SS, CAS, TS
Hu et al. [30]	2019	RCT	22	21	Senior	Periodontology	Practice course	KS, TS
Huang et al. [31]	2014	RCT	30	30	Master	Orthodontics	Practice course	SS
Yu et al. [32]	2018	RCT	30	30	Senior	Orthodontics	Theory course	KS, CAS, TS
Wei et al. [33]	2015	RCT	40	40	Senior	Prosthodontics	Theory course	KS, CAS, TS
Sun et al. [34]	2019	RCT	15	15	Senior	Endodontics	Practice course	KS, CAS, TS
Yang et al. [35]	2018	RCT	20	20	Senior	Periodontology	Practice course	KS, SS, TS
Li BH et al. [36]	2015	RCT	40	40	Senior	Oral and maxillofacial esthetic surgery	Theory course	KS, SS, CAS, TS
Du et al. [37]	2013	RCT	20	20	Senior	Oral Medicine	Theory course	KS

CAS: Comprehensive ability scores; CBL: Case-based learning; KS: Knowledge scores; LBL: Lecture-based learning; No.: Number; RCT: Randomized controlled trial; SS: Skill scores; TS: Teaching satisfactions.

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Table 2: Results of subgroup analyses in the current meta-analysis.

	No. of participants				Incidence			
Items	**Studies**	**CBL**	**LBL**	***P***	**SMD/RR**	**95%CI**	**Heterogeneity(I^2)**	**Model**
Knowledge scores *								
Theoretical courses	14	656	651	<0.0001	1.78	0.95-2.62	97%	Random
Practical courses	7	226	185	<0.0001	1.1	0.34-1.86	91%	Random
Skill scores *								
Theoretical courses	2	40	40	<0.0001	4.06	3.28-4.84	N. S	Random
Practical courses	6	212	209	0.02	0.83	0.15-1.51	90%	Random
Comprehensive ability scores *								
Theoretical courses	7	309	309	<0.0001	3.81	1.88-5.74	98%	Random
Practical courses	4	132	130	0.01	1.39	0.82-1.97	72%	Random
Teaching satisfactions **								
Theoretical causes	6	293	294	<0.0001	2.63	1.78-3.91	65%	Random
Practical courses	2	47	46	0.0001	6.31	2.47-16.13	N. S	Random

CBL: Case-based learning; CI: Confidence interval; LBL: Lecture-based learning; N.S: Non-significant; No.: Number; RR: Risk ratio; SMD: Standard mean difference; * Values presented as mean difference and 95% confidence interval; * * Values presented as risk ratio and 95% confidence interval.

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Table 3: Subgroup analyses of the level of training in the current meta-analysis.

	No. of participant				Incidence			
Items	**Studies**	**CBL**	**LBL**	***P***	**SMD/MD**	**95%CI**	**Heterogeneity (I^2)**	**Model**
Knowledge scores *								
Junior	4	263	231	0.11	0.82	-0.17-1.82	96%	Random
Senior	22	686	671	0.001	1.53	0.75-2.31	97%	Random
The fifth year student	3	82	82	0.007	1.05	0.29-1.81	81%	Random

CBL: Case-based learning; CI: Confidence interval; LBL: Lecture-based learning; No.: Number; MD: Mean difference; SMD: Standard mean difference; * Values presented as mean difference and 95% confidence interval.

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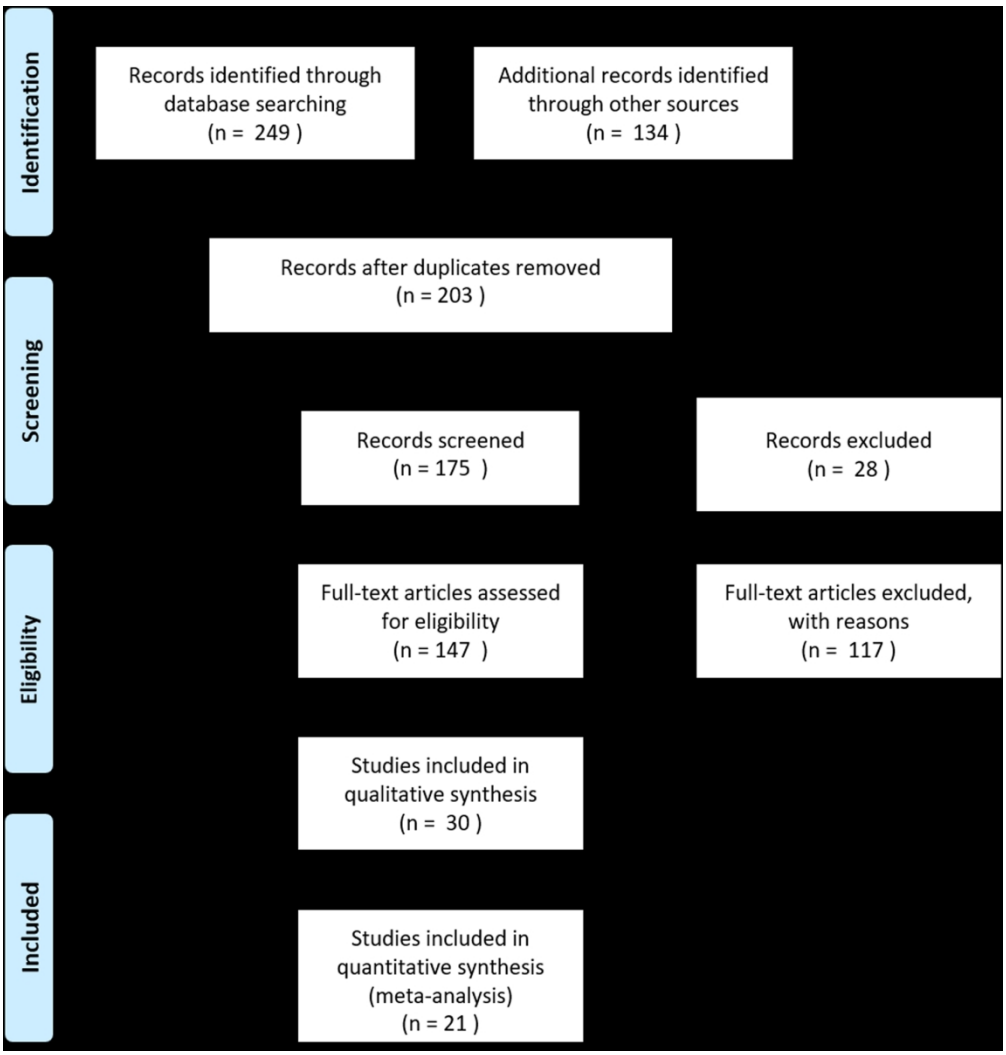


Figure 1: Flow diagram of study selection (preferred reporting items for systematic review and meta-analysis, PRISMA).

165x173mm (300 x 300 DPI)

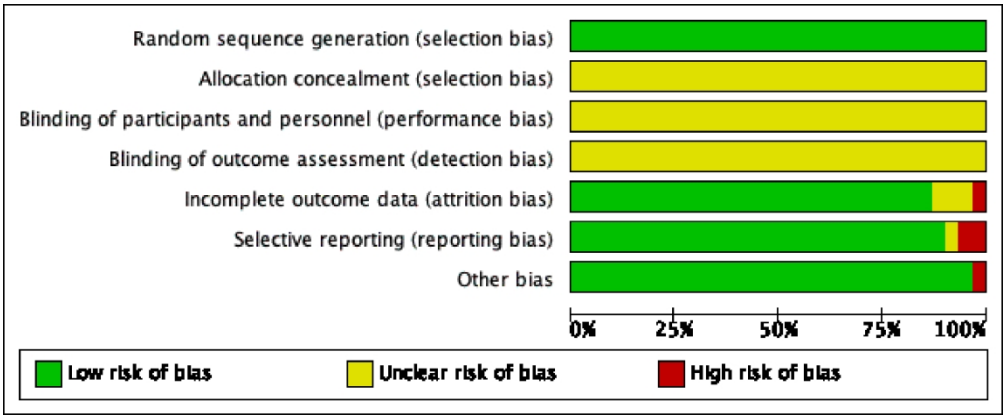


Figure 2: Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

219x90mm (300 x 300 DPI)

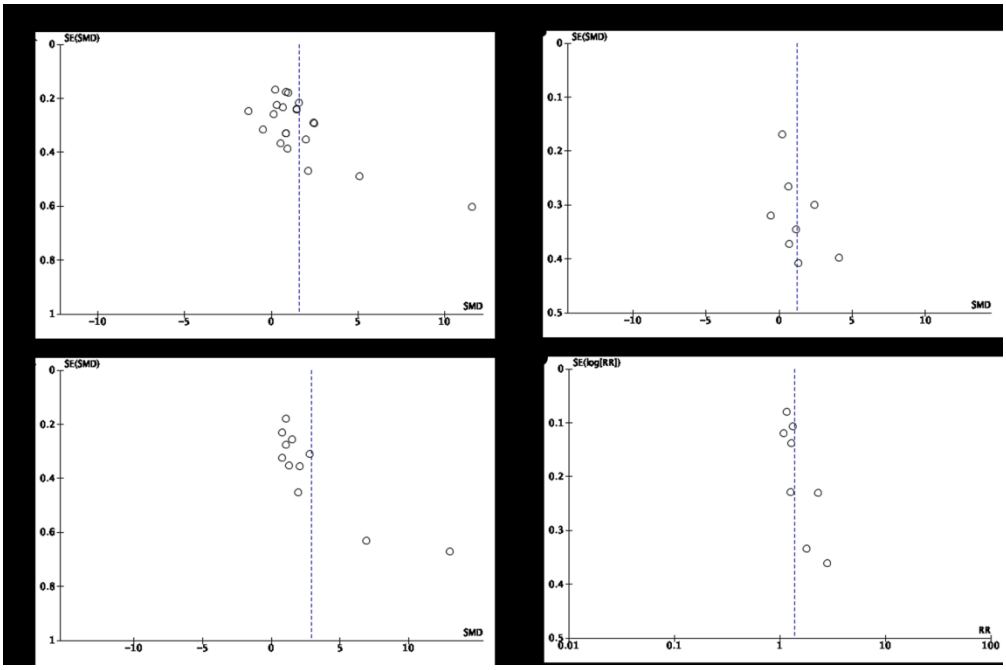


Figure 3: Funnel plots for publication bias. (A) Publication bias of knowledge scores. (B) Publication bias of skill scores. (C) Publication bias of comprehensive ability scores. (D) Publication bias of satisfactory teaching effect.

271x179mm (300 x 300 DPI)

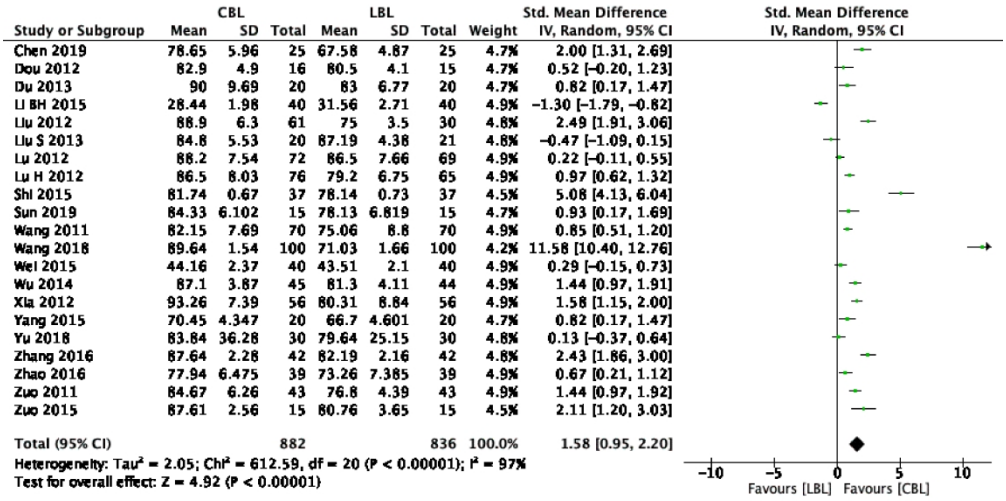


Figure 4: Forest plot of knowledge scores for CBL compared with LBL.

314x157mm (300 x 300 DPI)

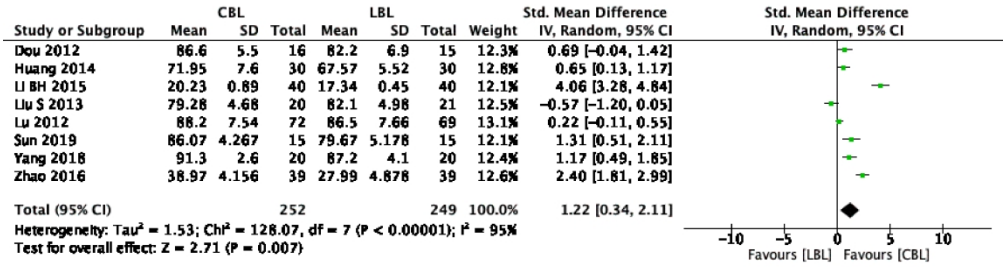


Figure 5: Forest plot of skill scores for CBL compared with LBL.

314x84mm (300 x 300 DPI)

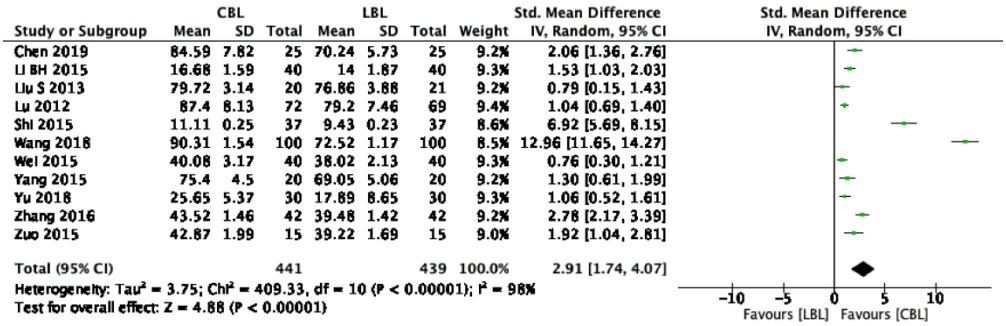


Figure 6: Forest plot of comprehensive ability scores for CBL compared with LBL.

308x101mm (300 x 300 DPI)

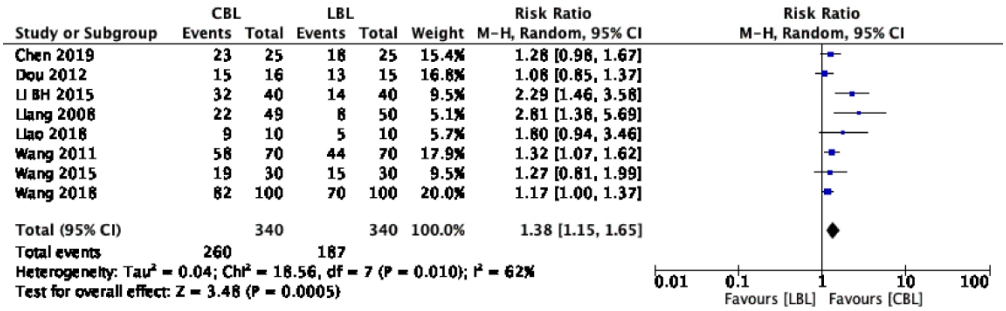


Figure 7: Forest plot of teaching satisfaction for CBL compared with LBL.

287x90mm (300 x 300 DPI)

PRISMA 2009 Checklist

Section/Topic	#	Checklist Item	Reported on Page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5-6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5-6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7

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PRISMA 2009 Checklist

Section/Topic	#	Checklist Item	Reported on Page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	9-10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-10
Synthesis of results	21	Present the main results of the review. If meta-analyses done, include for each, confidence intervals and measures of consistency.	8-10
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8-10
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	10
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	12
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	12
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097



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